An Explainable AI-Assisted Early Timing Violation Prediction Report

Open-Source EDA Integration • Generated: September 07, 2026 - 10:46:44

Design Module: mux_2to1	**Clock Constraint:** 0.500 ns (2000.0 MHz)
Prediction Verdict: TIMING CONSTRAINTS SATISFIED (NO VIOLATION)	**Confidence Score:** 96.5% (Prob: 0.965)

Synthesized Microarchitectural & Physical Layout Features

Parameter Feature	Value	Parameter Feature	Value
Design Name	mux_2to1	Clock Period	0.5
Clock Frequency	2000.0	Number of Standard Cells	4
Number of Flip-Flops	0	Number of Nets	16
Total Cell Area	12.0	Cell Utilization	0.40490000000000004
Maximum Fanout	14	Logic Depth	2
Placement Density	0.4	Routing Congestion	44.05
Estimated Wire Length	58.2	Buffer Count	0
Inverter Count	0	Estimated Critical Path Length	0.3598
TimingViolation	NO		

Automated VLSI Timing Closure Optimization Directives

Parameter	Status	Recommended Action Directive	Technical Rationale
Overall Design Constraints	**NORMAL**	**No Timing Optimization Required**	Design parameters satisfy setup timing margins without violation risks.

IEEE Model Verification & Feature Driver Plots



